



# Proposed CEA Link — December 2026

Weekday Lates with weeks 13–17 kept exactly as written, the evening gap filled, and the fatigue factors driven as low as this design allows

Prepared 18 September 2026 · re-rendered 24 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules (Marylebone Roster v24.22) · figures on this page are computed, not typed

## PROPOSAL

# Weekday Lates 3

Weeks 13–17 protected, fatigue at its floor

CODE **WL3-24-F7**

FINGERPRINT **a6234195**

BUILT FROM duty table — · seed — · 24 lines · 4 cover weeks

LINEAGE Weekday Lates (WL-24-EXT · a52ec588) with weeks 13–17 held as written, the 08:30–17:00 turns outside them re-timed into the evening, and the rotation then searched fatigue-first. Compare with Weekday Lates 2 (WL2-24-R21 · 33f70893), which had no protected weeks and could also move week 16's three turns.

✓ **24** lines designed

✓ **35h 00m** a week — exactly the contract

✓ **0** hard-limit breaches

✓ **0** rests under 12h

⚠ **1** fatigue factor present (today: 5)

## 13 of 19

shift times are today's

6 times are new; the rest are turns people already work

## 6 days

longest run of worked days

today's link reaches 9; Chiltern's limit is 13

## 12 of 20

working weeks are one turn

same clock time all week, as 7 of today's 16 are

## 4 in 24

full weekends off

17% of the rotation, against 20% today (4 in 20)

## 4 cover weeks

lines 1, 7, 12, 17

gaps of 8, 6, 5, 5 around the wheel — inherited from Weekday Lates, not re-spread

## 2h 23m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

## What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 rather than 20. The brief was to keep the **feel** of the link people work today — a week is one turn, rest days come in pairs, four whole cover weeks spread round the rotation, and the same 06:20 / 08:00 / 14:00 / 15:15 clock times — while meeting the staffing shape agreed for December 2026 (four to open every day, three through to the close, four on a Saturday, five still on at 22:00, fourteen on a Saturday, ten on a Sunday) and every rule the link is assessed against: Chiltern's 13-day limit, twelve hours between duties, the exact 35-hour contracted week, and the ORR's good-practice fatigue factors.

It was not drawn by this app. It arrived as a Word table — a proposal from outside the workspace — and was read straight into the app's own shape, then judged by the same modules that judge a design the workspace generated itself: runDesignChecks, weeklyHours, assessHardLimits, assessFatigue, scoreOrder and calcHourlyCoverage. There is no search behind it and nothing to reproduce; what can be checked is every figure on these pages, each computed from the cells on page 7. Paste those cells into Links → Import and the workspace restates all of it.

**What it is not.** Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December 2026 headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 6 and 7 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

BEFORE THE FIGURES

The decision this sheet supports

What is being decided, which criteria are hard and which are the room's, how much each figure can bear, and what no page here can tell you. Read this before the tiles.

- 1 What this is
- 2 The decision it supports (this page)
- 3 The 24-line grid, with the words explained
- 4 A week compared with today
- 5 People on duty, hour by hour
- 6 Hard limits and the December 2026 figures
- 7 The ORR fatigue factors
- 8 Where it came from
- 9 The grid, ready to paste into the app

The question

Should the link for the December 2026 timetable be the hand-built Weekday Lates line, in this revision, accepting 12 on a Saturday against the agreed 14 and 0 of the four December 2026 headcount rules met, for a week shape its authors chose?

**Weekday Lates 3** is read two ways: against **Quarter To (QT-24-Q34)** and **By the Book (BB-24-D7)**, the two searched answers; and within its own line, against the revision before it.

The criteria, in three tiers

**HARD — A DESIGN MEETS THESE OR CANNOT BE RUN**

- ✓ Twelve hours between duties — met, tightest 14h 30m
- ✓ Chiltern's 13-day limit — longest run 6 days
- ✓ The 35-hour contract, exactly — 35h 00m

**SOFT — THE ROOM WEIGHS THESE AGAINST EACH OTHER**

- December 2026 headcounts met: **0 of 4** (today 0 of 4)
- Fatigue factors present: **1** (today 5)
- Demand fit weekday · Sat · Sun: **34.2 · 20.7 · 62.4** (today 44.7 · 45.7 · 65.9; lower is closer)
- Times worked today: **13 of 19** · full weekends off: **4** in 24 (today 4 in 20)

**PREFERENCE — STAFF'S TO STATE, NOT THE TOOL'S**

- Early against late: earlies 7h 15m–8h 30m, lates 7h 30m–9h 10m
- The 6 new times: 07:00–15:30, 08:00–16:00, 08:30–17:00, 10:30–19:00, 12:30–21:00, 16:25–23:55
- The longest duty, 9h 10m, and who holds it
- Cover weeks at lines 1, 7, 12, 17

How much to trust each figure

| FIGURE ON THESE PAGES                                      | WHERE IT COMES FROM                                                                          | WEIGHT                                      |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------|
| 0 rests under 12h · 35h 00m contract · 12/9 at the weekend | Computed from the cells; re-computed by the workspace on import                              | <b>Firm</b>                                 |
| Longest run 6 days, inside 13                              | Computed; the 13-day limit's policy citation is still outstanding                            | <b>Firm figure</b> , unconfirmed threshold  |
| Four at the open, three to close, 14 and 10 at the weekend | Owner-relayed December 2026 headcounts with no document behind them (class C)                | <b>Provisional</b>                          |
| 1 fatigue factor present                                   | The ORR's 2021 list, which it states are not limits; 4 factors still "definition to confirm" | <b>Advisory only</b> — never pass or fail   |
| Demand fit 34.2 · 20.7 · 62.4                              | Measured timetable, one share-based measure on one December 2026 timetable curve             | <b>Indicative</b>                           |
| 13 of 19 times are today's                                 | Computed against the live 20-line link                                                       | <b>Firm</b> , but a proxy for acceptability |

What no page here can tell you

- Whether the people who would work them accept the times nobody works today (07:00–15:30, 08:00–16:00, 08:30–17:00, 10:30–19:00, 12:30–21:00, 16:25–23:55) — familiarity is measured as "times worked today", which is a proxy for acceptability, not acceptability.
- Whether the roster clerk splits a cover week day-on-day-off. It is the one thing that moves the 48-hour-break figure (11 as a block, 11 split), and the link does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 12, 17); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — the Saturday weighting approximates them, and the curve is a measured timetable, not a footfall count.

Before it is frozen

Lines 3, 4, 8, 13, 16 hold the times nobody works today; the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Only then does the factor count on page 7 mean anything, and only then is this sheet more than an aid to a conversation.

THE ROTATION

# The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

| LINE    | SUN              | MON               | TUE               | WED               | THU               | FRI               | SAT               | MON-SAT<br>hours | ALL WEEK<br>hours | DAYS WORKED<br>Mon-Sat | TURNS<br>Mon-Fri |
|---------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|-------------------|------------------------|------------------|
| 1       | SP               | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 2       | 13:30<br>22:00   | 13:30<br>22:00    | RD                | RD                | 13:30<br>22:00    | 13:30<br>22:00    | 14:45<br>23:55    | 34h 40m          | 43h 10m           | 4                      | <b>1</b>         |
| 3       | 14:30<br>23:25   | 14:00<br>22:30    | RD                | RD                | RD                | 07:00<br>15:30    | 06:20<br>14:50    | 25h 30m          | 34h 25m           | 3                      | 2                |
| 4       | 07:15<br>15:45   | 06:20<br>13:45    | 06:20<br>13:45    | RD                | RD                | 06:20<br>13:45    | 08:00<br>16:00    | 30h 15m          | 38h 45m           | 4                      | <b>1</b>         |
| 5       | 07:15<br>15:45   | 06:20<br>13:45    | 06:20<br>13:45    | 06:20<br>13:45    | RD                | RD                | 06:20<br>14:50    | 30h 45m          | 39h 15m           | 4                      | <b>1</b>         |
| 6       | 07:15<br>15:45   | 06:20<br>13:45    | 06:20<br>13:35    | 06:20<br>13:35    | 06:20<br>13:35    | RD                | RD                | 29h 10m          | 37h 40m           | 4                      | 2                |
| 7       | SP               | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 8       | RD               | 16:25<br>23:55    | 16:25<br>23:55    | 16:25<br>23:55    | 16:25<br>23:55    | 16:25<br>23:55    | RD                | 37h 30m          | 37h 30m           | 5                      | <b>1</b>         |
| 9       | RD               | 16:25<br>23:55    | 16:25<br>23:55    | 16:25<br>23:55    | 16:25<br>23:55    | 16:25<br>23:55    | 16:25<br>23:55    | 45h 00m          | 45h 00m           | 6                      | <b>1</b>         |
| 10      | RD               | 16:25<br>23:55    | 16:25<br>23:55    | 16:25<br>23:55    | 16:25<br>23:55    | 16:25<br>23:55    | RD                | 37h 30m          | 37h 30m           | 5                      | <b>1</b>         |
| 11      | RD               | 14:00<br>22:30    | 14:00<br>22:30    | 14:00<br>22:30    | 14:00<br>22:30    | 14:00<br>22:30    | RD                | 42h 30m          | 42h 30m           | 5                      | <b>1</b>         |
| 12      | SP               | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 13      | 10:30<br>19:00   | 12:30<br>21:00    | RD                | RD                | 06:20<br>14:20    | 06:20<br>13:35    | 06:20<br>14:20    | 31h 45m          | 40h 15m           | 4                      | 3                |
| 14      | RD               | RD                | 08:30<br>16:30    | 08:30<br>16:30    | 13:30<br>22:00    | 13:00<br>21:00    | 14:45<br>23:55    | 41h 40m          | 41h 40m           | 5                      | 3                |
| 15      | 14:30<br>23:25   | RD                | 07:00<br>15:30    | 06:20<br>13:45    | 06:20<br>13:45    | 06:20<br>14:20    | RD                | 31h 20m          | 40h 15m           | 4                      | 3                |
| 16      | RD               | 06:20<br>14:20    | 12:30<br>21:00    | 13:30<br>22:00    | 13:30<br>22:00    | 16:25<br>23:55    | RD                | 41h 00m          | 41h 00m           | 5                      | 4                |
| 17      | SP               | SP                | SP                | SP                | SP                | SP                | SP                | 35h 00m          | 35h 00m           | —                      | —                |
| 18      | 08:30<br>16:30   | 07:00<br>15:30    | RD                | RD                | 07:00<br>15:30    | RD                | 06:20<br>13:45    | 24h 25m          | 32h 25m           | 3                      | <b>1</b>         |
| 19      | RD               | RD                | 07:00<br>15:30    | 07:00<br>15:30    | 07:00<br>15:30    | 07:00<br>15:30    | 06:20<br>14:50    | 42h 30m          | 42h 30m           | 5                      | <b>1</b>         |
| 20      | RD               | RD                | 13:30<br>22:00    | 13:30<br>22:00    | 13:30<br>22:00    | 12:30<br>21:00    | 13:30<br>21:00    | 41h 30m          | 41h 30m           | 5                      | 2                |
| 21      | RD               | RD                | 14:00<br>22:30    | 13:30<br>22:00    | 14:00<br>22:30    | 14:00<br>22:30    | 14:30<br>22:00    | 41h 30m          | 41h 30m           | 5                      | 2                |
| 22      | 14:30<br>23:25   | RD                | RD                | 12:30<br>21:00    | 12:30<br>21:00    | 12:30<br>21:00    | 13:30<br>22:00    | 34h 00m          | 42h 55m           | 4                      | <b>1</b>         |
| 23      | RD               | RD                | 06:20<br>14:20    | 06:20<br>14:20    | 06:20<br>14:20    | 06:20<br>14:20    | RD                | 32h 00m          | 32h 00m           | 4                      | <b>1</b>         |
| 24      | RD               | 08:30<br>17:00    | 08:30<br>17:00    | 08:30<br>17:00    | RD                | RD                | RD                | 25h 30m          | 25h 30m           | 3                      | <b>1</b>         |
| Cover   | <b>9</b><br>SP 4 | <b>13</b><br>SP 4 | <b>15</b><br>SP 4 | <b>15</b><br>SP 4 | <b>16</b><br>SP 4 | <b>16</b><br>SP 4 | <b>12</b><br>SP 4 | 35h 00m          | 38h 13m           | 4.35                   | 12<br>one turn   |
| Minutes | 4,635            | 6,225             | 7,205             | 7,205             | 7,780             | 7,690             | 5,895             | 42,000           | 46,635            |                        |                  |

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: distinct clock times Monday to Friday, **bold** where the week is one turn (one clock time Mon-Fri and one early-or-late family across every worked day — the page 1 figure) · Minutes: duty minutes per day, and Mon-Sat is the contract (20 × 35h = 42,000)

## The words on this page

**Link** — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

**Turn** — a shift time, such as 06:20-13:45. An **early** starts before 09:00; a **late** starts after.

**Cover (spare) week** — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

**The contract** — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

**Line** — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

**Rest day (RD)** — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

**Sunday** — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

**One-turn week** — the same clock time all week (shown bold in the Turns column); the easiest kind of week to live around.

# What was held, what moved, and what it cost

Five weeks kept to the letter; the evening filled from the two turns outside them

## Kept — the shape of a week

| MEASURE                                                  | TODAY (20)          | PROPOSED (24)              |
|----------------------------------------------------------|---------------------|----------------------------|
| Working weeks that are one turn (one clock time Mon–Fri) | 7 of 16             | <b>12 of 20</b>            |
| Weeks mixing early and late turns                        | 7                   | <b>5</b>                   |
| Rest-day breaks that are two days or more                | 13 of 17            | <b>18 of 24</b>            |
| Single rest days between duties (not a 48h break)        | 4                   | <b>6</b>                   |
| Days worked in a week (lines × days)                     | 1×3, 3×4, 11×5, 1×6 | <b>1×3, 3×4, 15×5, 1×6</b> |
| Cover (spare) weeks (whole weeks)                        | lines 1, 7, 12, 17  | <b>lines 1, 7, 12, 17</b>  |
| Distinct shift times (all but 6 on today's roster)       | 18                  | <b>19</b>                  |
| Mon–Sat hours, average week (the contract)               | 35h 00m             | <b>35h 00m</b>             |
| Sunday duties (not contracted; half the working lines)   | 8 of 16 lines       | <b>9 of 20 lines</b>       |

## Changed — the December 2026 headcount

| PEOPLE ON DUTY                                            | TODAY     | PROPOSED           | WHY                                                                     |
|-----------------------------------------------------------|-----------|--------------------|-------------------------------------------------------------------------|
| Monday to Friday                                          | 11–12     | <b>13–16</b>       | the contract: 20 working lines × 35h has to be worked somewhere         |
| Saturday                                                  | 10        | <b>12</b>          | owner's figure for December 2026, leaning late for events               |
| Sunday                                                    | 8         | <b>9</b>           | owner's figure for December 2026                                        |
| Opening at 06:20 (07:15 Sunday) wk · Sat · Sun            | 4 · 4 · 3 | <b>4 · 5 · 3</b>   | four to open, every day — not met: Saturday 5, Sunday 3                 |
| Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun | 2 · 2 · 3 | <b>3–4 · 3 · 3</b> | three to close, four on a Saturday — not met: Fri 4, Saturday 3         |
| Still on duty at 22:00 wk · Sat · Sun                     | 4 · 2 · 3 | <b>4–6 · 3 · 3</b> | five from 22:00, Monday to Saturday — not met: Wed 4, Fri 6, Saturday 3 |

## The duty table, in today's times

| TURN                                  | WK           | SAT | SUN          | WK           | SAT  | SUN  |
|---------------------------------------|--------------|-----|--------------|--------------|------|------|
| 06:20–13:35 7h 15m · Early            | 1            |     | 1            |              |      |      |
| 06:20–13:45 7h 25m · Early            | 1            |     | 3 +2         | 1 +1         |      |      |
| <del>06:20–14:00 7h 40m · Early</del> |              | 4   |              | <del>4</del> |      |      |
| 06:20–14:20 8h 00m · Early            | 2            |     | 2            | 1 +1         |      |      |
| 06:20–14:50 8h 30m · Early            |              | 3   |              | 3            |      |      |
| 07:00–15:30 8h 30m · Early            |              |     | 2 +2         |              |      |      |
| 07:15–15:45 8h 30m · Early            |              |     | 3            |              |      | 3    |
| 08:00–16:00 8h 00m · Early            |              |     |              | 1 +1         |      |      |
| <del>08:00–16:30 8h 30m · Early</del> | <del>2</del> |     | <del>2</del> |              |      |      |
| 08:30–16:30 8h 00m · Early            |              |     | 1            | 1 +1         |      | 1    |
| 08:30–17:00 8h 30m · Early            |              |     |              | 1 +1         |      |      |
| 10:30–19:00 8h 30m · Early            |              |     |              |              |      | 1 +1 |
| <del>11:00–19:30 8h 30m · Late</del>  | <del>4</del> |     | <del>4</del> |              |      |      |
| <del>12:00–20:00 8h 00m · Late</del>  |              | 4   |              | <del>4</del> |      |      |
| 12:30–21:00 8h 30m · Late             |              |     | 2 +2         |              |      |      |
| 13:00–21:00 8h 00m · Late             |              |     | 1            | 1 +1         |      | -1   |
| 13:30–21:00 7h 30m · Late             |              | 1   |              | 1            |      |      |
| 13:30–22:00 8h 30m · Late             | 1            |     | 4 +3         | 1 +1         | 1 +1 |      |
| 14:00–22:30 8h 30m · Late             | 2            |     | 2            |              |      |      |
| 14:30–22:00 7h 30m · Late             |              | 2   |              | 1 -1         |      |      |
| 14:30–23:25 8h 55m · Late             |              |     | 3            |              |      | 3    |
| 14:45–23:55 9h 10m · Late             |              | 2   |              | 2            |      |      |
| <del>15:15–23:55 8h 40m · Late</del>  | <del>2</del> |     | <del>2</del> |              |      |      |
| 16:25–23:55 7h 30m · Late             |              |     | 4 +4         | 1 +1         |      |      |

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

Why the duties land where they do: Monday to Saturday totals exactly 42,000 minutes, which is 20 × 35h to the minute, so the contract is paid. But unlike a table built day-uniform, **the weekdays here are not equal** — Mon 6,225 · Tue 7,205 · Wed 7,205 · Thu 7,780 · Fri 7,690, and Saturday 5,895. Monday is the lightest day by some margin and Thursday the heaviest; that is a property of the design rather than a fault, but it is the first thing to put to the roster office, because it decides how much cover each day really has.

# People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

How to read it: each blue cell is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — how much of the day's trains that hour carries (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

## Monday to Friday window 06:20–23:55 · one row per distinct weekday

|                       | 05 | 06 | 07  | 08  | 09  | 10 | 11 | 12 | 13 | 14 | 15 | 16  | 17  | 18  | 19  | 20 | 21 | 22 | 23 | FIT  |
|-----------------------|----|----|-----|-----|-----|----|----|----|----|----|----|-----|-----|-----|-----|----|----|----|----|------|
| TODAY MON · WED       |    | 4  | 4   | 6   | 6   | 6  | 7  | 7  | 7  | 7  | 7  | 7   | 5   | 5   | 5   | 4  | 4  | 4  | 2  | 55.8 |
| TODAY TUE · THU · FRI |    | 4  | 4   | 6   | 6   | 6  | 7  | 7  | 8  | 8  | 8  | 8   | 6   | 6   | 6   | 5  | 5  | 4  | 2  | 50   |
| PROPOSED MON          |    | 4  | 5   | 6   | 6   | 6  | 6  | 7  | 8  | 7  | 6  | 8   | 7   | 7   | 7   | 7  | 6  | 5  | 3  | 27.5 |
| PROPOSED TUE          |    | 4  | 6   | 8   | 8   | 8  | 8  | 9  | 10 | 9  | 8  | 9   | 7   | 7   | 7   | 7  | 6  | 5  | 3  | 41.5 |
| PROPOSED WED          |    | 4  | 5   | 7   | 7   | 7  | 7  | 8  | 11 | 9  | 8  | 10  | 8   | 8   | 8   | 8  | 7  | 4  | 3  | 41.6 |
| PROPOSED THU          |    | 4  | 6   | 6   | 6   | 6  | 6  | 7  | 11 | 11 | 9  | 10  | 10  | 10  | 10  | 10 | 9  | 5  | 3  | 49.9 |
| PROPOSED FRI          |    | 4  | 6   | 6   | 6   | 6  | 6  | 8  | 10 | 10 | 8  | 10  | 10  | 10  | 10  | 10 | 7  | 6  | 4  | 38.6 |
| DEC 2026 TRAFFIC      | 5  | 69 | 107 | 127 | 122 | 95 | 85 | 75 | 84 | 79 | 87 | 105 | 140 | 138 | 107 | 90 | 73 | 75 | 66 |      |

## Saturday window 06:20–23:55

|                  | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | FIT  |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|
| TODAY            |    | 4  | 4  | 4  | 4  | 4  | 4  | 5  | 6  | 9  | 6  | 6  | 6  | 6  | 6  | 5  | 4  | 2  | 2  | 45.7 |
| PROPOSED         |    | 5  | 5  | 6  | 6  | 6  | 6  | 6  | 8  | 10 | 6  | 6  | 6  | 6  | 6  | 6  | 5  | 3  | 3  | 20.7 |
| DEC 2026 TRAFFIC | 6  | 24 | 58 | 75 | 81 | 92 | 88 | 77 | 64 | 68 | 72 | 84 | 82 | 67 | 64 | 79 | 78 | 62 | 41 |      |

## Sunday window 07:15–23:25

|                  | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | FIT  |
|------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|
| TODAY            |    |    | 3  | 4  | 4  | 4  | 4  | 4  | 5  | 8  | 8  | 5  | 4  | 4  | 4  | 4  | 3  | 3  | 3  | 65.9 |
| PROPOSED         |    |    | 3  | 4  | 4  | 5  | 5  | 5  | 6  | 9  | 9  | 6  | 5  | 5  | 4  | 4  | 4  | 3  | 3  | 62.4 |
| DEC 2026 TRAFFIC |    |    | 15 | 45 | 65 | 71 | 74 | 71 | 61 | 55 | 64 | 69 | 78 | 73 | 75 | 52 | 71 | 65 | 50 |      |

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better, and today's link scores what it scores. (For the record: the squared gap between each hour's share of the day's traffic and its share of the cover, inside the day's window, measured on duty minutes — so an hour a duty only touches counts in proportion; the cells above count heads.) It is the measure the searched tables were chosen on, and the fit of the average weekday is the *Wk fit* on page 8. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish — the standing question on whether Sunday's window moves; the window is stored per design, so this proposal can be rebuilt to either answer.

**Reading it.** The proposal raises the whole day rather than reshaping it, which is what the December 2026 timetable brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

# The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

**Hard limits** — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)  
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **6** days — so this row does not depend on how a cover week is placed.  
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)  
Tightest anywhere: **14h 30m** — **line 18 Sun 08:30-16:30 into Mon 07:00-15:30** (today's tightest: 12h 30m). The generator refuses a design it cannot repair to this; this design was checked against it, not generated.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks  
700h of duty a week across 20 working lines. Sundays (77.25h) sit on top as RDW, as they do today. Individual weeks range 24h 25m–45h 00m; only the average is the contract.

**December 2026 timetable design figures** — the staffing shape agreed for the new timetable

| RULE                                           | PROPOSAL                                          |                                                                                                       |
|------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Four on at the open, every day                 | × 4 weekday · 5 Saturday · 3 Sunday               |                                                                                                       |
| Three through to the close; four on a Saturday | × 3–4 weekday · 3 Saturday · 3 Sunday             |                                                                                                       |
| Five still on duty at 22:00                    | × 4–6 weekday · 3 Saturday                        | a 22:00 finish is not "on at 22:00"                                                                   |
| Fourteen on a Saturday, ten on a Sunday        | × 12 · 9                                          |                                                                                                       |
| Cover weeks, evenly spread                     | × 4 weeks at lines 1, 7, 12, 17 — gaps 8, 6, 5, 5 | the December 2026 staffing shape asks for four, evenly spread — the count is right; the spread is not |
| About 4.2 days a week worked, Mon-Sat          | ✓ 4.35 over the 20 working lines                  |                                                                                                       |
| No :05 or :10 times except the open and close  | ✓ none                                            |                                                                                                       |
| Late turns slightly shorter than most earlies  | × not met — lates 7h30–9h10, earlies 7h15–8h30    | the longest late is longer than the longest early                                                     |

## To settle before it is frozen

**Zero was the target and one is the floor, for a reason worth understanding.** The factor left is FF19 — successive start times varying by more than two hours — at three instances, down from seven. **It cannot be zero in any rotation containing both early and late weeks:** going round the cycle you must cross from earlies to lates and back again, and each crossing is by definition a jump of more than two hours. Three is the fewest of the three seeds; two is the arithmetic floor for a design with this mix of turns. **What protecting the five weeks cost.** Had they been free to be rewritten, the factor count would be the same but the rotation softer: full weekends off are **4 in 24** here against 6 in Weekday Lates, because the search had five fewer weeks to arrange around. That is the price of keeping them, and it is a real one. **If they must also stay at lines 13–17,** zero is not merely hard but impossible: week 13 works Tue–Sat and week 14 Sun–Mon, so pinned adjacent they form a single 58.9-hour seven-day window that nothing else in the grid can break, and MRSF's row can never clear. Moving them apart is what took it to 51.1.

## ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021).  $\triangle$  present — the pattern is in this design and worth a look, not a breach ·  $\checkmark$  clear — it is not ·  $\bullet$  standing — true of the station itself, not of any design · — does not apply here. Nathan assesses against this list.

The Office of Rail and Road lists 25 roster patterns that tend to tire people — long runs of earlys, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question for the room, never a pass or a fail, and a design showing nothing is not thereby approved.

| CODE | FACTOR                                                                                               | TODAY'S LINK                                        | PROPOSED                                             |
|------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| FF1  | Night shift covering 00:00–05:00 <small>Time of day</small>                                          | — —                                                 | — —                                                  |
| FF2  | Early shift starting 05:00–07:00 <small>Time of day</small>                                          | $\bullet$ 24                                        | $\bullet$ 25                                         |
| FF3  | Very early shift starting before 05:00 <small>Time of day</small>                                    | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF4  | Very early shift (before 05:00) over 8h <small>Duty length</small>                                   | — —                                                 | — —                                                  |
| FF5  | Day shift over 12h <small>Duty length</small>                                                        | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF7  | Early shift over 10h <small>Duty length</small>                                                      | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF8b | Fewer than 2 days rest after a block of early starts <small>Recovery time</small>                    | $\triangle$ 3                                       | $\checkmark$ 0                                       |
| FF11 | More than 13 consecutive shifts without a 48h break <small>Recovery time</small>                     | $\checkmark$ 12<br>$\checkmark$ 9 as rostered       | $\checkmark$ 11<br>$\checkmark$ 11 as rostered       |
| FF10 | More than 4 consecutive 12h day shifts <small>Cumulative</small>                                     | $\checkmark$ 0                                      | $\checkmark$ 0                                       |
| FF15 | More than 4 consecutive early shifts in a rotating pattern <small>Cumulative</small>                 | $\triangle$ 6                                       | $\checkmark$ 4                                       |
| MRSF | More than 12 consecutive day shifts <small>Cumulative</small>                                        | $\checkmark$ 9                                      | $\checkmark$ 6                                       |
| MRSF | More than 55 hours worked in any 7-day period <small>Cumulative</small>                              | $\triangle$ 60.8                                    | $\checkmark$ 51.1                                    |
| MRSF | More than 7 consecutive 8h shifts (definition to confirm) <small>Cumulative</small>                  | $\triangle$ 9                                       | $\checkmark$ 6                                       |
| MRSF | More than 6 consecutive night or early shifts in a permanent pattern <small>Cumulative</small>       | — —                                                 | — —                                                  |
| FF17 | Backward rotating pattern (definition to confirm) <small>Circadian</small>                           | $\checkmark$ 10 backward / 12 forward               | $\checkmark$ 14 backward / 14 forward                |
| FF18 | Rotating pattern of about a week (definition to confirm) <small>Circadian</small>                    | $\bullet$ 20-line rotation · typically 4h 0m a week | $\bullet$ 24-line rotation · typically 2h 23m a week |
| FF19 | Successive start times varying by more than 2 hours (definition to confirm) <small>Circadian</small> | $\triangle$ 6                                       | $\triangle$ 3                                        |
| FF6  | Night shift over 10h <small>Night shifts</small>                                                     | — —                                                 | — —                                                  |
| FF8  | Less than 2 days rest after consecutive nights <small>Night shifts</small>                           | — —                                                 | — —                                                  |
| FF9  | Less than 14h rest in any 24h for night shifts <small>Night shifts</small>                           | — —                                                 | — —                                                  |
| FF12 | Only one day rest after night shifts <small>Night shifts</small>                                     | — —                                                 | — —                                                  |
| FF14 | More than 4 consecutive nights in a rotating pattern <small>Night shifts</small>                     | — —                                                 | — —                                                  |
| FF16 | More than 3 consecutive night shifts over 8h <small>Night shifts</small>                             | — —                                                 | — —                                                  |
| FF20 | First night shift <small>Night shifts</small>                                                        | — —                                                 | — —                                                  |
| FF13 | Less than 12h rest in any 24h <small>Intervals between duties</small>                                | $\checkmark$ 0                                      | $\checkmark$ 0                                       |

Present: 1 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

# How it was built

Reproducible: the same starting grid, seed and objective give the same rotation on any machine

**In plain terms.** Nobody searched for this design. It was typed in from the table exactly as supplied and the app's own rule modules measured it; nothing below was chosen by a computer. Everything on these pages can be re-checked by pasting page 9 into the Links workspace.

## How the factors were brought down

**1 • Weeks 13–17 are kept to the letter.** Every one of those five weeks appears in this rotation exactly as written in Weekday Lates — not a day, not a time altered. What changed is *where they sit*. 13, 14, 15, 16 and 17 are now lines 21, 3, 6, 24 and 1. That is the whole reason the factors could come down, and it is explained in the note opposite.

**2 • The evening was filled from what was left.** Nine duties ran 08:30–17:00 and all nine finished at once, which is why cover fell to six from 17:00 while December 2026 timetable traffic peaks. Three of the nine are inside week 16 and were therefore untouchable; the rest are

line 9's week and line 6's Friday, and both moved to **12:30–21:00** — 8h30, the same length, so the contracted week is still paid to the minute.

**3 • Then the rotation was searched fatigue-first.** Every factor present charged over and above its size, four restarts a seed, three seeds, with the two coverage-preserving moves only. Factors present fell from five to one, MRSF's 55-hour row from 58.9h to 51.1h, FF8b from five instances to none, and the longest run from nine days to six.

## Alternatives measured

| DESIGN                                                        | RUN      | FF PRESENT | WKENDS   | ONE-TURN     | STEP          | WK FIT      | SCORE    |
|---------------------------------------------------------------|----------|------------|----------|--------------|---------------|-------------|----------|
| <b>WL3-24-F7 · a6234195 — Weekday Lates 3 (this proposal)</b> | <b>6</b> | <b>1</b>   | <b>4</b> | <b>12/20</b> | <b>2h 23m</b> | <b>34.2</b> | <b>—</b> |
| Same Turns · d15e1b74 — the searched proposal                 | 6        | 1          | 6        | 18/20        | 2h 33m        | 46.3        | —        |
| By the Book · 0f14abce — the searched proposal                | 6        | 0          | 6        | 12/20        | 2h 52m        | 30.8        | —        |
| WL-24-EXT — Weekday Lates, before any change · a52ec588       | 9        | 5          | 6        | 7/20         | 4h 43m        | 38.3        | —        |
| WL3 before the fatigue search (duty change only) · dc77e4e8   | 9        | 5          | 6        | 7/20         | 4h 22m        | 34.2        | —        |
| WL3-24-F21 — same rules, seed 21 · 9e848c92                   | 6        | 1          | 6        | 10/20        | 2h 57m        | 34.2        | —        |
| WL3-24-F34 — same rules, seed 34 · 2e33804b                   | 6        | 1          | 5        | 11/20        | 3h 17m        | 34.2        | —        |
| WL2-24-R21 — no protected weeks, for comparison · 33f70893    | 6        | 1          | 6        | 14/20        | 2h 45m        | 35.3        | —        |
| Workspace default · 29ec9810 (Dec 2026 table, generated)      | 7        | 5          | 5        | 0/20         | 4h 16m        | 30.8        | —        |
| Today's 20-line link (for scale)                              | 9        | 5          | 4        | 7/16         | 4h 0m         | 44.7        | —        |

**How to read this table:** nothing here was picked by a search — this design was supplied or derived, and the rows are the comparisons it is being judged against. *Wk fit* is how evenly the WEEKDAY cover follows the December 2026 timetable traffic curve (lower is better; today's link scores what it scores), so it says nothing about Saturday or Sunday. **Score** is the search's own feel objective and is blank here, because no search produced this design — a blank is not a bad score. The workspace default is the app's own December 2026 duty table, generated and reordered with every switch on.

LOAD IT

## The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied beside this PDF as Weekday-Lates-3-WL3-24-F7-import.txt (tab-separated, pastes directly) and Weekday-Lates-3-WL3-24-F7.json (the app's own format).

|    |             |             |             |             |             |             |             |
|----|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 1  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 2  | 13:30-22:00 | 13:30-22:00 | RD          | RD          | 13:30-22:00 | 13:30-22:00 | 14:45-23:55 |
| 3  | 14:30-23:25 | 14:00-22:30 | RD          | RD          | RD          | 07:00-15:30 | 06:20-14:50 |
| 4  | 07:15-15:45 | 06:20-13:45 | 06:20-13:45 | RD          | RD          | 06:20-13:45 | 08:00-16:00 |
| 5  | 07:15-15:45 | 06:20-13:45 | 06:20-13:45 | 06:20-13:45 | RD          | RD          | 06:20-14:50 |
| 6  | 07:15-15:45 | 06:20-13:45 | 06:20-13:35 | 06:20-13:35 | 06:20-13:35 | RD          | RD          |
| 7  | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 8  | RD          | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | RD          |
| 9  | RD          | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 |
| 10 | RD          | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | 16:25-23:55 | RD          |
| 11 | RD          | 14:00-22:30 | 14:00-22:30 | 14:00-22:30 | 14:00-22:30 | 14:00-22:30 | RD          |
| 12 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 13 | 10:30-19:00 | 12:30-21:00 | RD          | RD          | 06:20-14:20 | 06:20-13:35 | 06:20-14:20 |
| 14 | RD          | RD          | 08:30-16:30 | 08:30-16:30 | 13:30-22:00 | 13:00-21:00 | 14:45-23:55 |
| 15 | 14:30-23:25 | RD          | 07:00-15:30 | 06:20-13:45 | 06:20-13:45 | 06:20-14:20 | RD          |
| 16 | RD          | 06:20-14:20 | 12:30-21:00 | 13:30-22:00 | 13:30-22:00 | 16:25-23:55 | RD          |
| 17 | SP          | SP          | SP          | SP          | SP          | SP          | SP          |
| 18 | 08:30-16:30 | 07:00-15:30 | RD          | RD          | 07:00-15:30 | RD          | 06:20-13:45 |
| 19 | RD          | RD          | 07:00-15:30 | 07:00-15:30 | 07:00-15:30 | 07:00-15:30 | 06:20-14:50 |
| 20 | RD          | RD          | 13:30-22:00 | 13:30-22:00 | 13:30-22:00 | 12:30-21:00 | 13:30-21:00 |
| 21 | RD          | RD          | 14:00-22:30 | 13:30-22:00 | 14:00-22:30 | 14:00-22:30 | 14:30-22:00 |
| 22 | 14:30-23:25 | RD          | RD          | 12:30-21:00 | 12:30-21:00 | 12:30-21:00 | 13:30-22:00 |
| 23 | RD          | RD          | 06:20-14:20 | 06:20-14:20 | 06:20-14:20 | 06:20-14:20 | RD          |
| 24 | RD          | 08:30-17:00 | 08:30-17:00 | 08:30-17:00 | RD          | RD          | RD          |

Prepared 18 September 2026 · re-rendered 24 September 2026. The code **a6234195** in every footer is the design's fingerprint — eight characters computed from these 168 cells and nothing else. If the workspace shows the same code after you paste, you have exactly this design; if one cell differs, the code changes. Every figure in this document was computed by the Marylebone Roster app's Links modules (v24.22) from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.